RAJASTHAN ROYALS · SupeRR SELECTOR SEASON 2

Identifying the Most Reliable Players Under Pressure

Round 1 Methodology Report

A reproducible, delivery-level framework that measures how batters and bowlers perform in high-pressure situations relative to their own baseline, across IPL and T20I cricket (2022–2026).

Eligible pool: 104 batters, 162 bowlers · 867 matches · 200,961 deliveries analysed · IPL + T20I combined, format-adjusted baselines

Confidential — prepared on supplied proprietary competition data. Not for distribution.

1 · Executive Summary

We define **pressure** as a continuous, per-delivery intensity (0–1) built from the match situation *before* each ball is bowled — innings phase, wickets already lost, balls remaining and, in run-chases, the required run rate and how alive the contest is. Crucially we build **two separate, asymmetric models**: a batting team is under pressure in hard chases and at the death, whereas a bowling team is under pressure when batters are set and scoring freely. Every player is then judged against **their own baseline** and scored on a balanced blend of (i) absolute pressure performance, (ii) uplift versus baseline, and (iii) match-to-match consistency, with empirical-Bayes shrinkage so small samples cannot spike. Rankings are stress-tested across nine model configurations; the **reliability** column is the share in which a player stays in the top 10.

Top 10 Batters Under Pressure

#	Batter	Score	Rel.	Pr. balls	Pr. mts	Pressure SR	Base SR	SR upl.	Pr. avg
1	**Matthew Wade**	88.3	67%	229	19	162	111	+51	53
2	**Namboori Tilak Varma**	85.8	89%	886	63	169	129	+40	39
3	**Naman Dhir**	84.4	100%	267	25	193	148	+45	32
4	**Rishabh Pant**	83.0	100%	339	31	182	128	+54	33
5	**Suryakumar Yadav**	82.9	89%	590	39	198	158	+40	40
6	**Sikandar Raza**	82.7	89%	357	26	164	120	+44	28
7	**Harry Tector**	81.4	78%	182	16	162	111	+51	30
8	**Shubman Gill**	81.3	56%	202	17	193	146	+47	39
9	**Tristan Stubbs**	81.2	78%	613	49	168	125	+44	38
10	**Virat Kohli**	77.4	22%	172	15	189	142	+47	54

Top 10 Bowlers Under Pressure

#	Bowler	Type	Score	Rel.	Pr. balls	Pr. mts	Pr. wkts	Pressure econ	Base econ	Wkts/ over
1	**Craig Young**	Pace	91.5	78%	98	11	9	8.45	8.69	0.55
2	**Wanindu Hasaranga**	Spin	88.5	100%	432	37	33	7.79	7.95	0.46
3	**Spencer Johnson**	Pace	87.9	44%	97	14	8	9.34	9.20	0.49
4	**Azmatullah Omarzai**	Pace	85.3	78%	131	19	9	8.93	8.67	0.41
5	**Vaibhav Arora**	Pace	84.3	78%	171	22	15	10.14	9.61	0.53
6	**Mitchell Starc**	Pace	84.1	89%	171	20	19	10.11	9.49	0.67
7	**Ravi Bishnoi**	Spin	83.7	67%	223	28	18	9.09	8.22	0.48
8	**Digvesh Rathi**	Spin	82.9	44%	146	14	9	8.26	9.30	0.37
9	**Mukesh Choudhary**	Pace	78.8	56%	97	13	7	9.40	9.62	0.43
10	**Kartik Tyagi**	Pace	77.8	44%	115	12	6	9.34	10.70	0.31

Rel. = reliability (of 9 configs). Pr. = pressure. SR upl. = pressure SR minus own baseline SR. Wkts credited to the bowler only (run-outs excluded).

2 · Problem Statement & Approach

Round 1 asks us to build a data-driven system that measures how batters and bowlers perform in high-pressure situations compared with their usual level, and to rank the players who deliver most consistently. The definition of pressure is left to us. Our guiding principles were:

- **Reliability over reputation.** Every pick is derived from the supplied ball-by-ball data; no player list is hard-coded.
- **Self-referenced.** A high pressure strike rate is only credited relative to that batter's own baseline — we reward players who *maintain or raise* their level when it matters.
- **No future leakage.** Pressure at a delivery uses only information available up to that ball; the match result is never used to label earlier deliveries.
- **Honest about the data.** Where a quantity (e.g. exact team totals) cannot be reconstructed reliably, we say so and avoid building on it.
- **Robustness first.** A single formula is not trusted; the answer must survive reasonable changes to thresholds, weights and the pressure definition itself.

3 · Dataset

Seven supplied files. The core is **ball_by_ball.csv** (200,961 main-innings deliveries after excluding 6 super-over innings), joined to match metadata (883 matches, 867 with ball data), competitions (141; 5 IPL seasons 2022–2026), teams, playing XIs and competition–team–player mappings. Coverage splits roughly evenly: ~397 T20 matches (371 IPL) and ~470 T20I. All joins verified; the delivery key (match, innings, ball) is unique.

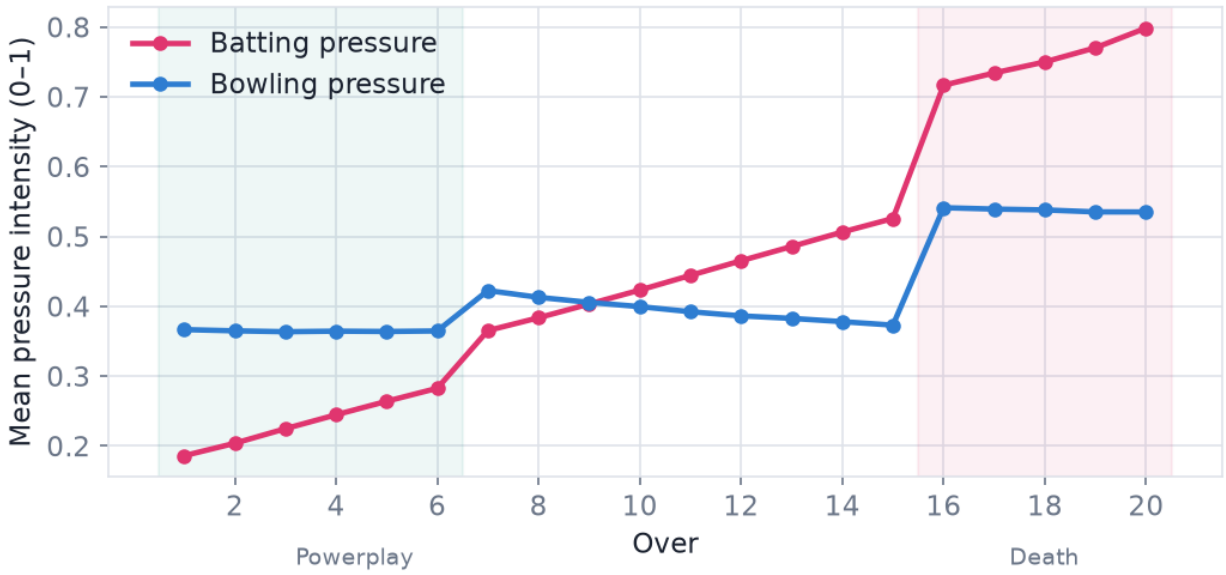
4 · Data Cleaning & the Extras Puzzle

The single most important data-quality finding: **extras1_runs is not a number of runs — it is a category code**, one-to-one with extras_type (6 = legal ball, 2 = wide, 3 = leg-bye, 4 = bye, 1 = no-ball, 5 = other, 7 = penalty). Summing it would be meaningless. We confirmed that **runs** is **runs off the bat** (every six has runs = 6, every four runs = 4, with zero contradictions; all wides/byes/leg-byes carry runs = 0). Therefore:

- **Batting metrics are exact** — strike rate, boundary %, dot %, dismissals all come straight from reliable fields.
- **Bowler economy is reliable** — byes and leg-byes are not charged to the bowler, so a bowler's runs = off-bat + wides + no-balls (+ recorded extra runs).
- **Team totals / required run rate are approximate** — bye/leg-bye magnitudes beyond 1 are not stored. Validated against 356 result margins: median error 0, 266 within ±3 runs; the large misses are all DLS rain games, which we exclude from chase logic.
- **Bowler wickets** exclude run-outs, retired-out and obstruction (609 of 10,881); 10,272 are credited. **Legal balls** exclude wides, no-balls and the re-bowled ‘other’ extra. All aggregation keys on player_id, since some names map to two IDs.

5 · Pressure Definition

Pressure is a weighted blend of normalised signals, each in [0,1], combined differently for batting and bowling. The chart below shows the mean intensity by over: batting pressure climbs steadily and jumps at the death; bowling pressure is flatter but peaks when batters attack — exactly the asymmetry we want.



Batting pressure rises with

- **Innings phase** — death overs carry the highest stake, powerplay the lowest.
- **Wickets already lost** — a rebuild after a collapse is higher pressure.
- **Required run rate** (chases) — a demanding chase is high pressure, but a *hopeless* chase (needing >3 runs/ball with few resources) is dampened, since garbage-time hitting is not a fair clutch test.

Bowling pressure rises with

- **Death overs**, where margins are smallest.
- **Batting momentum** — a high current run rate with wickets in hand means the bowler must stem the flow.
- **Defending a gettable chase** — when the batting side is cruising (low required rate), the bowler is under most pressure to break through.

A delivery is classified ‘pressure’ when its intensity ≥ 0.55 (configurable; swept in §12). This yields 33% batting-pressure and 19% bowling-pressure deliveries.

6 · Match-State Reconstruction

For every delivery we reconstruct the state before the ball, chronologically: score so far, wickets fallen, wickets in hand, legal balls bowled and remaining, and — in the second innings — the target (first-innings total + 1), runs required, required run rate and current run rate. Balls remaining are clamped to ≥ 0; DLS matches are flagged and excluded from chase-based pressure. Correctness is covered by unit tests on hand-built innings (score, wickets, legal-ball counting, RRR, and a no-future-leakage invariant).

7 · Batter Performance Metrics

On each delivery set (pressure and the player's own baseline) we compute: balls faced, runs, strike rate, average (runs per dismissal, counting only the batter's own dismissals), dot-ball %, boundary %, and match counts. The headline comparison is pressure vs baseline strike rate and survival.

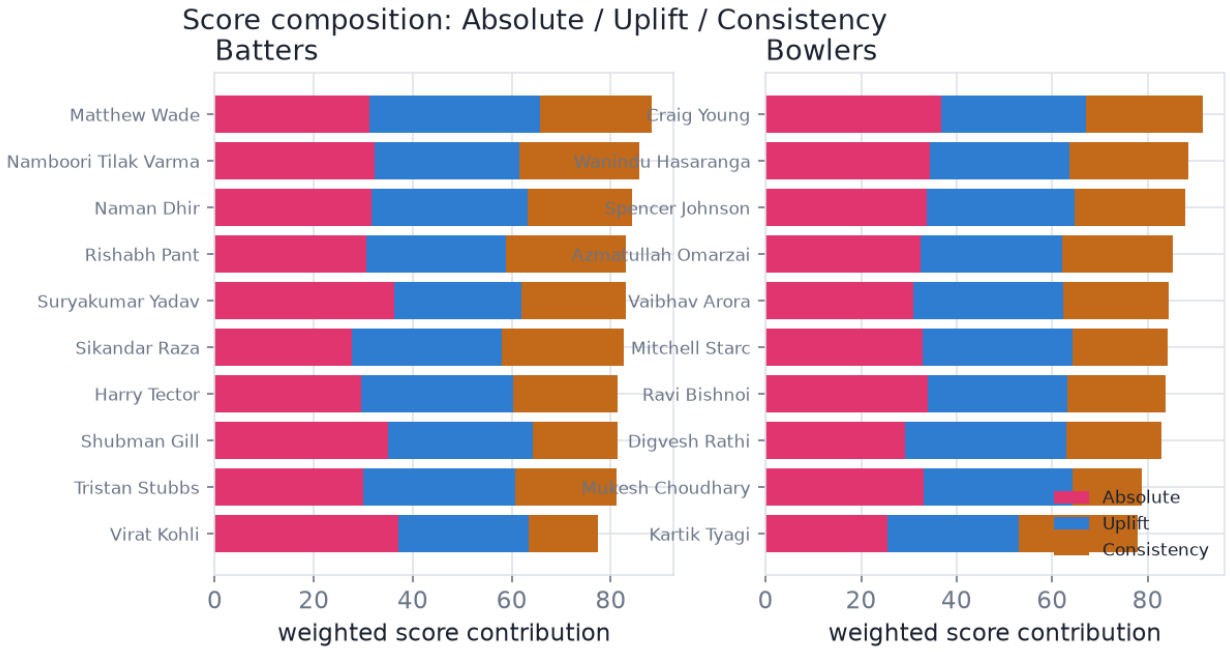
8 · Bowler Performance Metrics

Per delivery set: legal balls, runs conceded (bowler-charged), credited wickets, economy, strike rate, wicket rate per over, dot-ball % and boundary % conceded — each compared against the bowler's own baseline.

9 · Player Rating Methodology

Score = 0.40 · Absolute + 0.35 · Uplift + 0.25 · Consistency (each a 0–1 percentile within the eligible pool, ×100):

- **Absolute** — performance under pressure, *format-adjusted* against the league's pressure par for the formats the player actually played (so IPL and T20I are compared fairly). Batters: pressure SR + survival; bowlers: economy-vs-par + wicket rate + dot %.
- **Uplift** — pressure minus own baseline, **empirical-Bayes shrunk** toward zero by pressure sample size, so a player with 15 good balls cannot outrank one with 300.
- **Consistency** — the share of a player's pressure *matches* in which they actually delivered (batter SR ≥ own baseline; bowler economy ≤ baseline or took a wicket), reliability-weighted by number of qualifying matches.



10 · Sample Size, Eligibility & Reliability

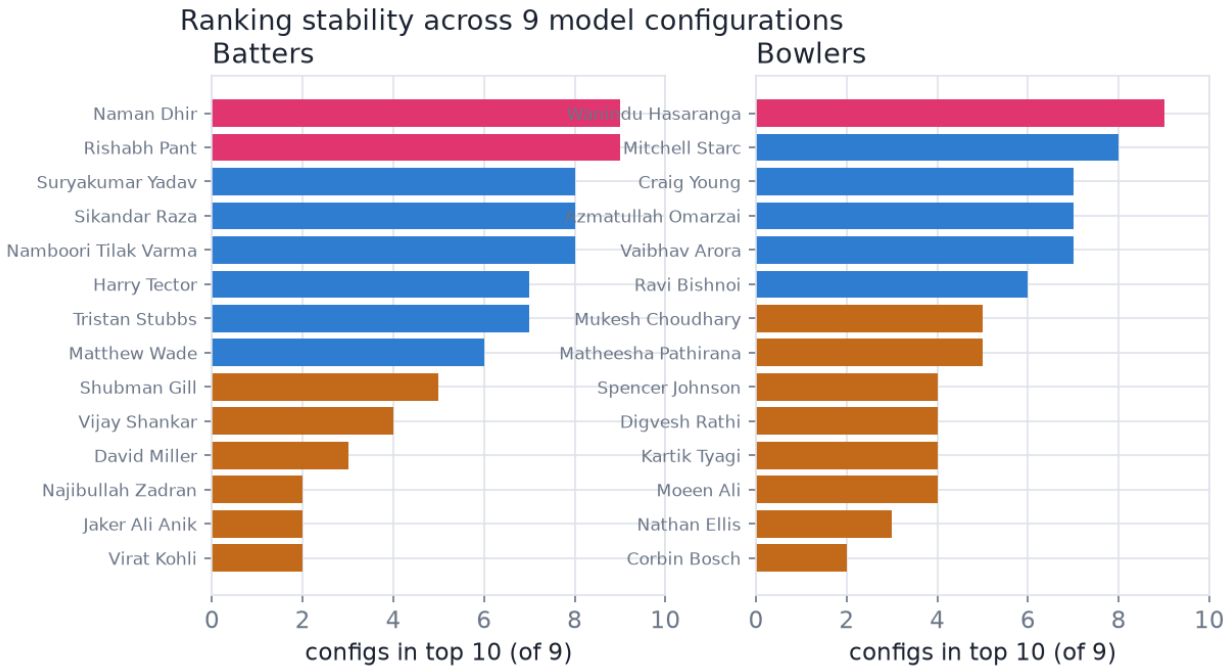
Eligibility (configurable): batters need ≥ 75 pressure balls, ≥ 5 pressure matches, ≥ 150 baseline balls; bowlers ≥ 60 pressure legal balls, ≥ 5 pressure matches, ≥ 120 baseline balls. Shrinkage priors (60 balls for batters, 48 for bowlers) pull small-sample uplifts toward zero. We never tune thresholds to force a particular player in.

11 · IPL and T20I Comparison

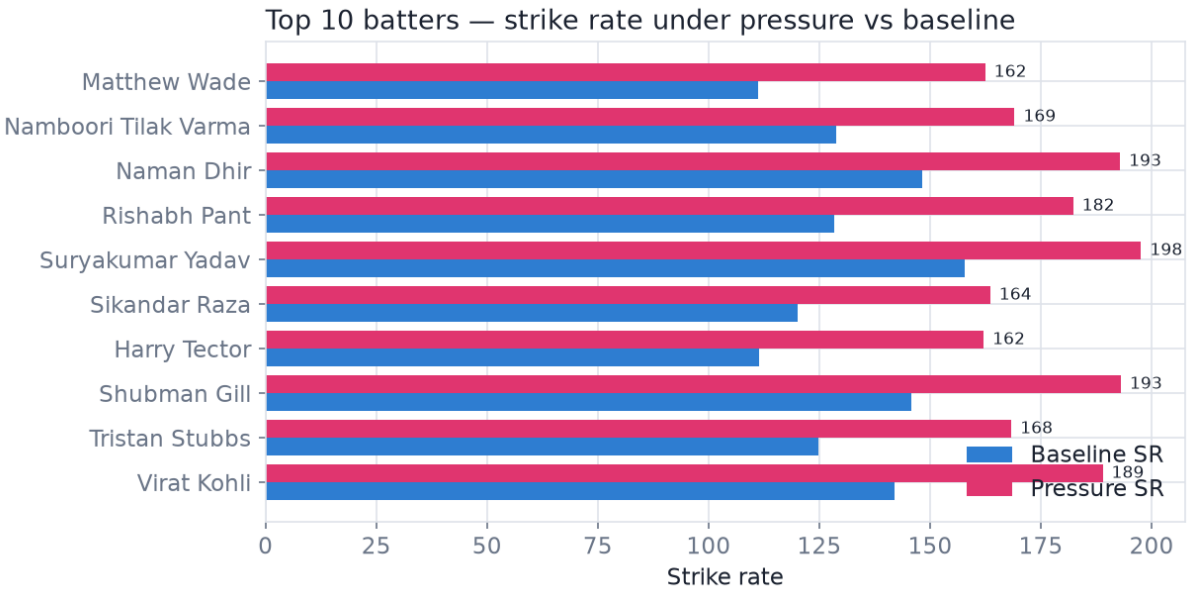
Both formats are combined, but baselines are computed per player (their own non-pressure balls in the formats they played) and the absolute component is adjusted against format-specific pressure par (IPL pressure SR ≈ 153 vs T20I ≈ 138; IPL pressure economy ≈ 10.6 vs T20I ≈ 10.0). Two sensitivity configs — IPL-weighting and a strict-sample cut — check that associate-nation inflation does not drive the picks.

12 · Sensitivity & Ranking Stability

We re-ran the entire rating under nine reasonable configurations: three pressure thresholds (0.50/0.55/0.60), two alternative batting definitions (chase-centric; death-and-collapse with no RRR) and two bowling definitions (death-heavy; defend-heavy), a strict-sample variant, and three weight mixes (uplift-, absolute- and consistency-heavy). The bars show how often each player stayed in the top 10.


Ranking stability across 9 model configurations
Batters
Naman Dhir
Rishabh Pant
Suryakumar Yadav
Sikandar Raza
Namboori Tilak Varma
Harry Tector
Tristan Stubbs
Matthew Wade
Shubman Gill
Vijay Shankar
David Miller
Najibullah Zadran
Jaker Ali Anik
Virat Kohli
0
2
4
6
8
10
configs in top 10 (of 9)
Bowlers
Wanindu Hasaranga
Mitchell Starc
Craig Young
Azmatullah Omarzai
Vaibhav Arora
Ravi Bishnoi
Mukesh Choudhary
Matheesha Pathirana
Spencer Johnson
Digvesh Rathi
Kartik Tyagi
Moeen Ali
Nathan Ellis
Corbin Bosch
0
2
4
6
8
10
configs in top 10 (of 9)

13 · Top 10 Batters — Evidence



#	Batter	Score	Rel.	Pr. balls	Pr. mts	Pressure SR	Base SR	SR upl.	Pr. avg
1	**Matthew Wade**	88.3	67%	229	19	162	111	+51	53
2	**Namboori Tilak Varma**	85.8	89%	886	63	169	129	+40	39
3	**Naman Dhir**	84.4	100%	267	25	193	148	+45	32
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5	**Suryakumar Yadav**	82.9	89%	590	39	198	158	+40	40
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8	**Shubman Gill**	81.3	56%	202	17	193	146	+47	39
9	**Tristan Stubbs**	81.2	78%	613	49	168	125	+44	38
10	**Virat Kohli**	77.4	22%	172	15	189	142	+47	54

Why each batter is here (all figures from the supplied data)

1. Matthew Wade — pressure strike rate **162** vs baseline 111 (**+51**), averaging 53 under pressure over 229 pressure balls in 19 matches; delivered in 81% of qualifying pressure matches; stays top-10 in 67% of configurations.

2. Namboori Tilak Varma — pressure strike rate **169** vs baseline 129 (**+40**), averaging 39 under pressure over 886 pressure balls in 63 matches; delivered in 75% of qualifying pressure matches; stays top-10 in 89% of configurations.

3. Naman Dhir — pressure strike rate **193** vs baseline 148 (**+45**), averaging 32 under pressure over 267 pressure balls in 25 matches; delivered in 75% of qualifying pressure matches; stays top-10 in 100% of configurations.

4. Rishabh Pant — pressure strike rate **182** vs baseline 128 (**+54**), averaging 33 under pressure over 339 pressure balls in 31 matches; delivered in 82% of qualifying pressure matches; stays top-10 in 100% of configurations.

5. Suryakumar Yadav — pressure strike rate **198** vs baseline 158 (**+40**), averaging 40 under pressure over 590 pressure balls in 39 matches; delivered in 69% of qualifying pressure matches; stays top-10 in 89% of configurations.

6. Sikandar Raza — pressure strike rate **164** vs baseline 120 (**+44**), averaging 28 under pressure over 357 pressure balls in 26 matches; delivered in 83% of qualifying pressure matches; stays top-10 in 89% of configurations.

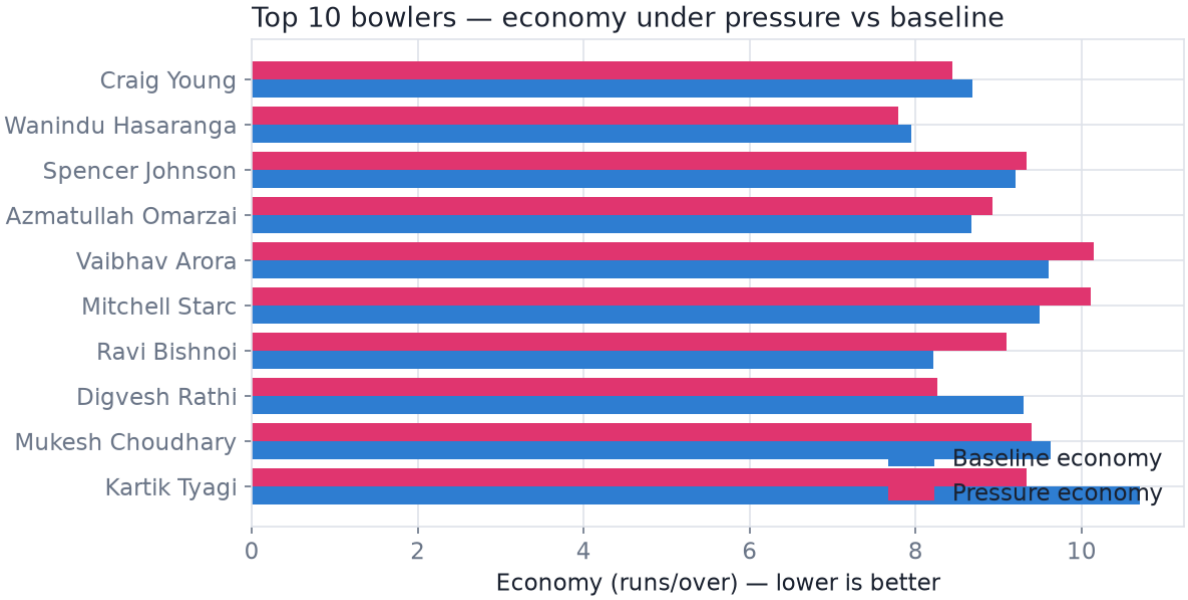
7. Harry Tector — pressure strike rate **162** vs baseline 111 (**+51**), averaging 30 under pressure over 182 pressure balls in 16 matches; delivered in 83% of qualifying pressure matches; stays top-10 in 78% of configurations.

8. Shubman Gill — pressure strike rate **193** vs baseline 146 (**+47**), averaging 39 under pressure over 202 pressure balls in 17 matches; delivered in 73% of qualifying pressure matches; stays top-10 in 56% of configurations.

9. Tristan Stubbs — pressure strike rate **168** vs baseline 125 (**+44**), averaging 38 under pressure over 613 pressure balls in 49 matches; delivered in 68% of qualifying pressure matches; stays top-10 in 78% of configurations.

10. Virat Kohli — pressure strike rate **189** vs baseline 142 (**+47**), averaging 54 under pressure over 172 pressure balls in 15 matches; delivered in 73% of qualifying pressure matches; stays top-10 in 22% of configurations.

14 · Top 10 Bowlers — Evidence



#	Bowler	Type	Score	Rel.	Pr. balls	Pr. mts	Pr. wkts	Pressure econ	Base econ	Wkts/ over
1	**Craig Young**	Pace	91.5	78%	98	11	9	8.45	8.69	0.55
2	**Wanindu Hasaranga**	Spin	88.5	100%	432	37	33	7.79	7.95	0.46
3	**Spencer Johnson**	Pace	87.9	44%	97	14	8	9.34	9.20	0.49
4	**Azmatullah Omarzai**	Pace	85.3	78%	131	19	9	8.93	8.67	0.41
5	**Vaibhav Arora**	Pace	84.3	78%	171	22	15	10.14	9.61	0.53
6	**Mitchell Starc**	Pace	84.1	89%	171	20	19	10.11	9.49	0.67
7	**Ravi Bishnoi**	Spin	83.7	67%	223	28	18	9.09	8.22	0.48
8	**Digvesh Rathi**	Spin	82.9	44%	146	14	9	8.26	9.30	0.37
9	**Mukesh Choudhary**	Pace	78.8	56%	97	13	7	9.40	9.62	0.43
10	**Kartik Tyagi**	Pace	77.8	44%	115	12	6	9.34	10.70	0.31

Why each bowler is here (all figures from the supplied data)

1. Craig Young (pace) — economy **8.45** vs baseline 8.69, 0.55 credited wickets/over and 41% dot balls under pressure, over 98 pressure legal balls in 11 matches; stays top-10 in 78% of configurations.

2. Wanindu Hasaranga (spin) — economy **7.79** vs baseline 7.95, 0.46 credited wickets/over and 34% dot balls under pressure, over 432 pressure legal balls in 37 matches; stays top-10 in 100% of configurations.

3. Spencer Johnson (pace) — economy 9.34 (vs 9.20 baseline), 0.49 credited wickets/over and 41% dot balls under pressure, over 97 pressure legal balls in 14 matches; stays top-10 in 44% of configurations.

4. Azmatullah Omarzai (pace) — economy 8.93 (vs 8.67 baseline), 0.41 credited wickets/over and 40% dot balls under pressure, over 131 pressure legal balls in 19 matches; stays top-10 in 78% of configurations.

5. Vaibhav Arora (pace) — economy 10.14 (vs 9.61 baseline), 0.53 credited wickets/over and 36% dot balls under pressure, over 171 pressure legal balls in 22 matches; stays top-10 in 78% of configurations.

6. Mitchell Starc (pace) — economy 10.11 (vs 9.49 baseline), 0.67 credited wickets/over and 40% dot balls under pressure, over 171 pressure legal balls in 20 matches; stays top-10 in 89% of configurations.

7. Ravi Bishnoi (spin) — economy 9.09 (vs 8.22 baseline), 0.48 credited wickets/over and 39% dot balls under pressure, over 223 pressure legal balls in 28 matches; stays top-10 in 67% of configurations.

8. Digvesh Rathi (spin) — economy **8.26** vs baseline 9.30, 0.37 credited wickets/over and 32% dot balls under pressure, over 146 pressure legal balls in 14 matches; stays top-10 in 44% of configurations.

9. Mukesh Choudhary (pace) — economy **9.40** vs baseline 9.62, 0.43 credited wickets/over and 41% dot balls under pressure, over 97 pressure legal balls in 13 matches; stays top-10 in 56% of configurations.

10. Kartik Tyagi (pace) — economy **9.34** vs baseline 10.70, 0.31 credited wickets/over and 35% dot balls under pressure, over 115 pressure legal balls in 12 matches; stays top-10 in 44% of configurations.

15 · IPL-Availability Filter

Because Rajasthan Royals is an IPL franchise, the final lists exclude players who cannot be signed in the IPL — specifically players whose national team (inferred from their T20I appearances in the data) is Pakistan. The scoring model is unchanged and fully data-driven; this is an actionability filter applied *after* scoring, and the removed players remain in the full metrics tables for transparency.

Excluded batters (would otherwise be eligible): Babar Azam, Mohammad Rizwan, Fakhar Zaman, Usman Khan, Iftikhar Ahmed, Salman Agha, Hasan Nawaz, Mohammad Haris.

Excluded bowlers: Haris Rauf, Abbas Afridi, Sufiyan Muqeem, Abrar Ahmed, Naseem Shah, Shadab Khan, Shaheen Afridi, Mohammad Nawaz, Mohammad Wasim.

The filter is a single configurable setting; removing it reinstates these players in rank order.

16 · Model Validation

- **Unit-tested cricket logic** — hand-built innings verify score/wicket reconstruction, legal-ball counting, bowler wicket credit, required run rate, and a no-future-leakage invariant (11/11 pass).
- **Innings totals cross-checked** against 356 result margins (median error 0; 266 within ±3 runs; large misses are DLS games).
- **Pressure definition is meaningful** — mean intensity rises monotonically through the innings and differs by role, matching cricketing intuition.
- **Small samples are shrunk, not rewarded** — the sample-size-vs-score plots show no runaway effect from low-ball players.
- **Rankings are stable** — the strongest picks appear in the top 10 across nearly all nine configurations.

17 · Limitations

- **Approximate targets** — bye/leg-bye magnitudes are not stored, so required run rate is accurate to ±1–3 runs; DLS matches are excluded from chase logic. Coarse pressure bands make this immaterial to classification.
- **Pressure is a modelling choice**, not an official RR definition; we mitigate with nine configurations and per-player reliability.
- **T20I strength varies** by opponent; baselines are format-adjusted but some associate-nation inflation can remain — checked via IPL-weighted and strict-sample configs.
- **No fitted win-probability model** — contest closeness is proxied by required run rate, wickets and phase.
- **Availability filter** uses nationality inferred from T20I appearances; a handful of uncapped players have no T20I record and are treated as IPL-eligible.

18 · Conclusion

The framework turns the vague idea of ‘clutch’ into a reproducible, delivery-level measurement: a transparent pressure index, self-referenced performance, a balanced and shrinkage-protected rating, and a robustness check across nine model variants. The resulting top-10 batters and bowlers are not reputation picks — every name is backed by its own pressure-versus-baseline numbers and a reliability score, and every figure in this report is traceable to the supplied data via the accompanying code (run_pipeline.py) and interactive dashboard.
